

DATABRICKS

Responsible AI Hackathon

Thursday 9 July 2026

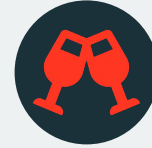
From use case to hands-on build — designed, governed, and delivered responsibly.

Housekeeping



Lunch

Served from 12:30 — please help yourself in the [breakout area] before we kick off.



Drinks & refreshments

Tea, coffee and soft drinks available throughout the day at [refreshment point].



WiFi

Network and password details are on [screen / table cards] — ask a facilitator if you need help.



Facilities & fire exits

Toilets are [location]; fire exits are clearly marked — please familiarise yourself on arrival.

TODAY'S PLAN

Agenda



09:30	●	Welcome	15 min
09:45	●	Use case identification session	1 hr
10:45	●	Intro to Databricks + hands-on exercises	1 hr
11:45	●	Intro to hackathon problem statements	30 min
12:30	●	Lunch	
13:15	●	Hacking	Rest of the day

Timings are indicative, based on the session durations agreed for today.



KICKING THINGS OFF

Welcome from RSS



SESSION

Value Mapping with Manuka.AI



GETTING SET UP

Vocareum & Databricks Intro

Introduction to Concepts



Lakehouse & Unity Catalog

A single governed home for data, features and models.



Agent Bricks (ai_query / ai_extract)

LLM building blocks for extraction and classification.



Human-in-the-loop review

A person checks AI output before it drives a decision.



Responsible AI Playbook

Principles (P1–P10) that guide safe, fair, governed use.



Fairness & bias checks

Tools like Fairlearn to test for disparate impact.

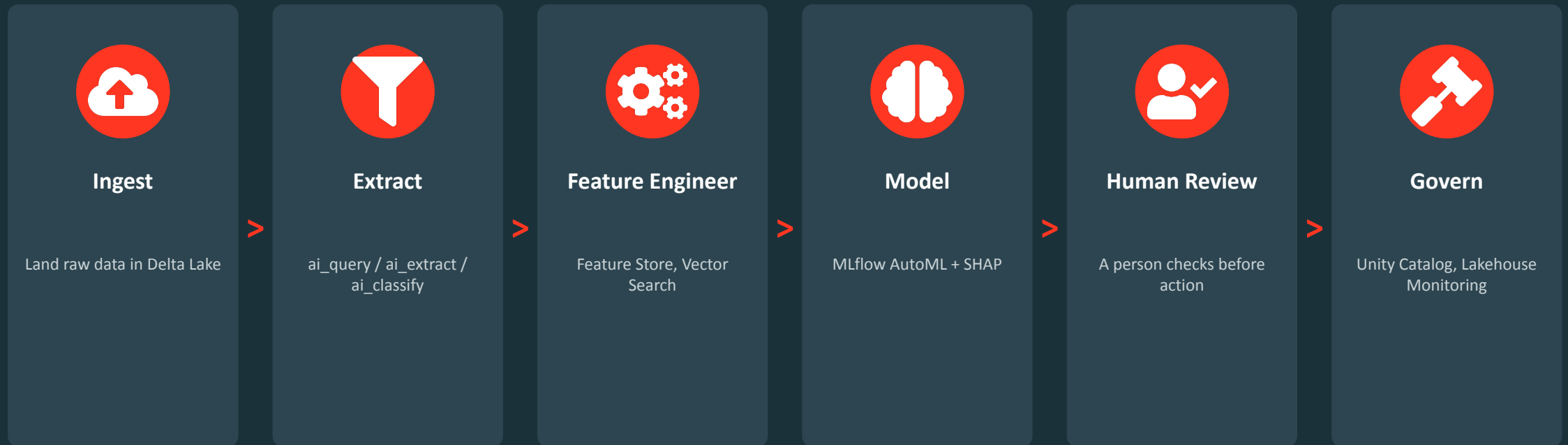


Explainability

SHAP and similar tools to show why a model decided something.

Every problem follows the same pipeline

Each problem statement moves through the same six stages — only the data and tools change.



What each team will do



1. Pick a problem statement

Choose one of the four use cases and agree your team's scope.



2. Work the pipeline

Ingest, extract, feature engineer and model using the suggested Databricks tools.



3. Apply the Playbook

Address the Responsible AI Playbook principles flagged for your problem.



4. Tackle a stretch goal

If time allows, go further using the stretch goal ideas on your problem card.



5. Be ready to discuss

Come prepared to answer the discussion questions on your problem card.



PROBLEM STATEMENT 1

CQC Inspection Report Analysis

Surfacing systemic risk in adult social care from inspection narratives

CONTEXT

CQC publishes inspection reports for every regulated care service in England, scored across five key questions (Safe, Effective, Caring, Responsive, Well-led). There's currently no scalable way to spot patterns in the weaknesses flagged across underperforming services — by geography, provider type, size, or time.

ANALYTICAL GOAL

Extract structured findings from report narratives, compare across geography and provider type, and surface early indicators of systemic risk for a human analyst to act on.

STRETCH GOAL

Build a geographic risk heatmap and detect emerging themes beyond current CQC categories.

DISCUSS

Where does the LLM confidently extract the wrong thing — and how would you catch that in production?

DATA

- CQC Open API — pre-loaded to Unity Catalog
- Care homes for the elderly, all English regions
- Open Government Licence — unrestricted use

DATABRICKS TOOLS

- ai_query / ai_extract
- Feature Store, Vector Search
- MLflow AutoML + SHAP
- Databricks Apps
- Fairlearn
- Unity Catalog & Lakehouse Monitoring

PLAYBOOK PRINCIPLES

P1 Know LLM extraction limits

P2 Bias check by region

P4 Analyst reviews before use

P6 OGL provenance via UC



PROBLEM STATEMENT 2

Social Unrest Early Warning

Building a responsible risk signal from news and social media text

CONTEXT

Home Office, DLUHC and Cabinet Office all have an interest in early warning signals for social unrest. Current approaches are largely reactive and manual — and building this kind of pipeline raises real questions about civil liberties, proportionality, and the scope of government monitoring.

ANALYTICAL GOAL

Ingest news and social text, extract semantic features (sentiment, entities, event type, intensity), and produce a monitorable risk signal with human review before any operational use.

STRETCH GOAL

Detect coordinated amplification with graph-based anomaly detection; alert when the risk index crosses a threshold.

DISCUSS

Where does the boundary sit between legitimate monitoring and surveillance in your pipeline?

Ethics requirement: pre-register your pipeline's scope (geographies, event types, platforms) and document what's out of scope, and why, before modelling.

DATA

- GDELT Project — daily global news events
- Reddit (Pushshift archive) — optional
- ONS economic indicators — optional context

DATABRICKS TOOLS

- Lakeflow Declarative Pipelines
- ai_query / ai_classify
- PySpark, topic modelling, Vector Search
- MLflow + SHAP
- Genie Spaces, Databricks Apps

PLAYBOOK PRINCIPLES

P1 LLMs can hallucinate event details

P2 Civil liberties — scope limits required

P4 Human triage before operational use

P10 ATRS registration if deployed



PROBLEM STATEMENT 3

Financial Fraud Detection

Combining structured and semantic features to improve anomaly detection

CONTEXT

HMRC, DWP, FCA and the Home Office all have a stake in fraud detection. Rule-based detectors work on structured fields, but fraud signals also live in unstructured text. LLM-derived features can enrich these models — but a false positive has real consequences for the person or business flagged.

ANALYTICAL GOAL

Combine structured transaction features with LLM-extracted semantic features to improve fraud classification, and show the model's reasoning is interpretable by an investigator.

STRETCH GOAL

Add transaction-graph features (PyG on Spark) and test whether semantic features shift false positive rates across demographic groups.

DISCUSS

A model that's 95% accurate sounds impressive — what does that mean if only 1% of transactions are actually fraudulent?

DATA

- IEEE-CIS Fraud Detection (Kaggle) — primary
- Synthetic transaction narratives
- BASD-8 / BABD-13 crypto graph data (LLM4TG, arXiv 2501.18158)

DATABRICKS TOOLS

- ai_extract / ai_classify
- Feature Store
- MLflow AutoML + SHAP
- Fairlearn
- Databricks Apps
- Unity Catalog & Model Registry

PLAYBOOK PRINCIPLES

P1 False positive risk is real

P2 Protected characteristics must not drive scores

P4 Investigator reviews before action

P10 Model card + ATRS required



PROBLEM STATEMENT 4

Financial Market Manipulation Detection

Fusing price signals and social media semantics to detect coordinated manipulation

CONTEXT

Market manipulation is increasingly organised via social media, and the FCA and Bank of England have a regulatory interest in tools that combine market data with online discourse. The GameStop/AMC episodes show how hard it is to distinguish manipulation from genuine retail enthusiasm.

ANALYTICAL GOAL

Combine price/volume anomaly detection with LLM-extracted social signals to identify potential manipulation, calibrated to a regulatory — not just a price-movement — definition.

STRETCH GOAL

Detect coordinated amplification via network analysis on posting patterns, and produce a precision/recall curve against a specific FCA definition.

DISCUSS

Retail investors have a right to discuss stocks publicly — where's the line between discussion and manipulation in your feature set?

Citation check: use LLM4TG (arXiv 2501.18158) for crypto graph analysis — not arXiv 2406.09012, which is unrelated (linguistic reference games).

DATA

- Reddit WallStreetBets (Pushshift) — primary text
- Yahoo Finance API — price & volume
- FinancialPhraseBank — sentiment baseline
- GDELT — corroborating news context

DATABRICKS TOOLS

- ai_classify / ai_query
- Delta Lake event storage
- PySpark (Granger causality, BSADF)
- MLflow + SHAP
- Databricks Apps
- Lakehouse Monitoring

PLAYBOOK PRINCIPLES

P1 LLMs misread irony / slang

P2 Data protection for individuals

P3 Calibrate to regulatory definition

P4 Compliance officer reviews before referral

AFTER TODAY

Next Steps



DEADLINE

Friday 17 July 2026

Email a copy of your notebook, slide deck, and video link to your Databricks contact.

WHAT TO SUBMIT



Notebook

A copy of the notebook your team worked in.



Slide deck

No more than 5 slides summarising your approach.



Video link

A recording where your team discusses what you did.

VIRTUAL SHOWCASE



5 minutes per team

Give an overview of your approach and what you built.



Live feedback

The panel will give feedback on your approach and outcome.



Winner announced

A winning team will be announced at the end of the showcase.



Stuck? Here's who to ask



General questions

Search or post on the Databricks Community — a good first stop for platform and tooling questions.

community.databricks.com



Vocareum issues

If you run into problems specifically with Vocareum access or your lab environment, email:

josh.bramall@databricks.com

